Arrays : Store multiple values of same type
(Basic Data Structure)

Declaring an array
Syntax : Data_type name [size];
It has a fixed size.

Indexing in an array is used to access the value stored at any index.

Array indexing starts from 0 to n - 1 , where n is the size of the array

int marks[5] = {5,3,4,2,1};
Marks[0] = 5
Marks[1] = 3
Marks[2] = 4
Marks[3] = 2
Marks[4] = 1

Vector : resizable array like structure.
It is convenient when size is unknown

Syntax :

vector<data_type> name(size);
E.g. vector<int> arr(5);

```
// Size of an array marks
Int n = marks.size();

// Iteration over indexing
int n = marks.size();
for(int i = 0; i < n; i++){
        cout << marks[i] << " ";
}

// Updating values in an array by user
for(int i = 0; i < n; i++){
         cin >> marks[i]; // i = 1, second input
      }

// Traversing an array : visiting every element in an
array/vector
for(int i = 0; i < n; i++){
         cout << marks[i] << " ";
      }


// Problem : sum of an array
      int sum = 0;
      // n = 5
      for(int i = 0; i  < n; i++){
```

```
        sum = sum + a[i]; // 180 + a[4] // 180 + 80
        // sum = 260
    }
    cout << sum ;
```

// Insert an element
Syntax : name.insert(a.begin() + position, x);

// Reverse an array or vector
swap(a, b);

a -> b
b -> a
Swap function swaps the values of passed two parameters!!

// while loop
Syntax :
Initialise
While(condition){
    // block of code
    I++; / i— ; // increment / decrement
  If we miss this step of increment and decrement , then our code or program will run for infinite time!!!

}

Homework :

Problem 1 : Write a program to find the sum of an array

Problem 2 : Reverse an array